THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · EASY LEVEL FULL-YEAR PAPER SET 2

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. If 2 : 3 = 8 : x, then x is: 1

(a) 8 **(b)** 12 **(c)** 14 **(d)** 16

2. 20% written as a fraction in lowest terms is: 1

(a) $\frac{25}{100}$ **(b)** 20 **(c)** $\frac{100}{20}$ **(d)** $\frac{1}{5}$

3. The hypotenuse of a right triangle is 5 cm and one leg is 3 cm. The other leg (in cm) is: 1

(a) 2 **(b)** 3 **(c)** 5 **(d)** 4

4. If 2:3 = 8:x, then x is: 1

(a) 16 **(b)** 7 **(c)** 12 **(d)** 11

5. The number of faces of a triangular prism is: 1

(a) 3 **(b)** 5 **(c)** 4 **(d)** 6

6. The median of 28, 19, 30, 34, 21, 38, 8 is: 1

(a) 34 **(b)** 8 **(c)** 25.43 **(d)** 28

7. The value of $3x + 4$ when $x = 5$ is: 1

(a) 21 **(b)** 17 **(c)** 19 **(d)** 38

8. The area of a triangle with base 10 cm and height 6 cm is: 1

(a) 30 cm² **(b)** 16 cm² **(c)** 60 cm² **(d)** 40 cm²

9. $\sqrt{169}$ is equal to: 1

(a) 15 **(b)** 14 **(c)** 13 **(d)** 12

10. $(2^3)^2$ equals: 1

(a) 2^9 **(b)** 2^7 **(c)** 2^5 **(d)** 2^6

11. The decimal value of 101_2 is: 1

(a) 9 **(b)** 5 **(c)** 3 **(d)** 7

12. In a parallelogram ABCD, $\angle A = 58^\circ$. Then $\angle C$ is: 1

(a) 302° **(b)** 90° **(c)** 58° **(d)** 122°

13. Which of the following is divisible by 2? 1

(a) 3851 **(b)** 3847 **(c)** 3849 **(d)** 3846

14. Expanding $3(x + 4)$ gives: 1

(a) $3x + 4$ **(b)** $x + 12$ **(c)** $3x + 12$ **(d)** $3x + 7$

15. In which of the following are the two ratios in proportion? 1

(a) $2 : 3$ and $9 : 12$ **(b)** $2 : 3$ and $12 : 8$

(c) $2 : 3$ and $8 : 12$ **(d)** $2 : 3$ and $8 : 13$

16. 10% of 250 is: 1

(a) 28 **(b)** 50 **(c)** 22 **(d)** 25

17. Which of the following is a Pythagorean triplet? 1

(a) 3, 4, 6 **(b)** 4, 5, 5 **(c)** 3, 4, 5 **(d)** 3, 5, 5

18. ₹ 400 is divided between two friends in the ratio 3:5. The larger share (in ₹) is: 1

(a) 250 **(b)** 270 **(c)** 150 **(d)** 200

19. The number of vertices of a triangular pyramid is: 1

(a) 4 **(b)** 6 **(c)** 3 **(d)** 5

20. The mode of 18, 16, 20, 18, 18, 9, 8 is: 1

(a) 18 **(b)** 16 **(c)** 8 **(d)** 20

SECTION — B (Very Short Answer Type) — 6 × 2 = 12

Answer in about 20–30 words each.

21. Write the ratio of 40 cm to 2 m in simplest form. 2

22. Express $\frac{3}{8}$ as a percentage. 2

23. The legs of a right triangle are 3 cm and 4 cm. Find its hypotenuse. 2

24. If 6 pens cost ₹ 90, find the cost of 15 pens. 2

25. Write the number of faces, vertices and edges of a triangular prism. 2

26. Find the mean of 38, 45, 19, 18, 20. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. A ladder 5 m long leans against a wall. Its foot is 3 m from the wall. How high on the wall does it reach? 3

28. The scale of a map is 1 cm = 12 km. Two cities are 5 cm apart on the map. Find the actual distance. Also find the map distance of two towns that are actually 96 km apart. 3

29. State the number of faces, vertices and edges of a triangular pyramid and verify Euler's formula. 3

30. The marks of 6 students are 42, 43, 48, 80, 34, 23. Find the mean mark and the number of students who scored above the mean. 3

31. Evaluate $3p + 4q - 5$ for $p = 6$, $q = 2$. 3

OR

Find the area of a parallelogram with base 14 cm and height 9 cm. What is the area of the triangle formed by half of it (cut along a diagonal)? 3

32. Find $\sqrt[3]{512}$ by prime factorisation. 3

OR

Write 4.2×10^{5} and 3.5×10^{-3} in usual form. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. The data of 30 students' choices is: Cricket 12, Football 9, Tennis 6, Hockey 3. Find the central angle of each sector for a pie chart and verify that the angles add to 360°. 5

34. A Koch curve begins with a straight line of length 9 cm (stage 0). At each stage every segment is replaced by 4 segments each one-third as long. Find the number of segments, the length of each segment and the total length at stage 2. 5

35. For the data 6, 25, 25, 4, 23, 8, find the mean, median, mode and range. 5

36. Think of a number n. Multiply it by 3, add 5 and then subtract 5. Show the result algebraically. What is the result when $n = 4$? 5

OR

A square plot has a perimeter of 24 m. Find its side and area. How many square tiles of side 1 m are needed to cover it? 5

37. Find the smallest number by which 108 must be divided so that the quotient is a perfect cube. Also find the cube root of the quotient. 5

OR

Simplify $(2^{2})^{3} \div 2^{4} \times 2^{1}$ and find its value. 5

38. Convert 19 to binary by repeated division by 2 and show the steps. 5

OR

ABCD is a rhombus of side 7 cm with $\angle A = 66^{\circ}$. Find $\angle B$, $\angle C$, $\angle D$ and the perimeter. 5

END OF QUESTION PAPER — Total: 80 Marks

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